

ADITYA TAMBE

Data Scientist | AI Engineer

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PROFESSIONAL SUMMARY

Data Scientist and AI Engineer with 7+ years of experience delivering end-to-end AI, Generative AI, and analytics solutions. Expertise in LLMs, RAG, Agentic AI, Machine Learning, Databricks, and PySpark, with deep domain experience in Pharma and Healthcare. Proven track record of translating business requirements into scalable, production-grade AI products.

TECHNICAL SKILLS

GenAI and LLMs: RAG, LangChain, LangGraph, LangSmith, LangServe, Ollama, OpenAI and Anthropic APIs, prompt engineering, fine-tuning, vector databases, Hugging Face

Machine Learning: Predictive modelling, time-series forecasting, classification and regression algorithms, XGBoost, Random Forest, Marketing Mix Modeling (MMx), NER

Programming: Python, PySpark, SQL, Pandas, NumPy, scikit-learn, web scraping (BeautifulSoup, Selenium)

Platforms and Tools: Databricks, AWS, Docker, Git, Flask, FastAPI, Streamlit, React, Power BI

PROFESSIONAL EXPERIENCE

IQVIA — Senior Programmer, Data Science

Mar 2025 – Present

Clinical trial data products for a leading global pharmaceutical client.

GenAI-Driven SAS-to-PySpark Migration for Clinical Trial Reporting

- Built a multi-agent GenAI system to analyze legacy SAS code, generate Databricks PySpark equivalents, perform automated code review, and validate clinical-trial reporting workflows.
- Designed specialized agents using LLM reasoning, structured prompts, tool execution, and clinical-domain logic across DM, AE, CM, Labs, and Vitals domains.
- Developed an automated SAS-to-PySpark parity agent to validate schemas, record counts, nulls, derived variables, and aggregate outputs against legacy results.
- Implemented a feedback-driven self-correction loop to identify discrepancies, refine generated code, and improve migration accuracy while reducing manual conversion effort.

Execution Monitoring Web Application with GenAI Failure-Diagnostics Chatbot

- Developed and deployed an end-to-end web application on Databricks to monitor pipeline executions across multiple studies and roles (IT, Clinical Programming, DRM).
- Implemented role-based access control aligned with study workflows so each user views only the executions relevant to their role.
- Integrated an LLM-powered chatbot for root-cause Q&A on failed jobs, reducing dependency on support teams for first-line triage.

Axtria — Senior Associate, Data Science

Jun 2023 – Feb 2025

Marketing Mix Modeling for Pharma

- Developed Marketing Mix Models (MMM) using Ridge/Elastic Net regression to quantify the incremental impact of promotional channels—including sales-rep calls, digital, email, and conferences—on prescription uptake and brand performance.
- Engineered adstock and saturation features to capture promotional carryover effects and diminishing returns across marketing channels.
- Quantified channel-level contribution and ROI, identifying high-performing channels and optimal spend levels for promotional investments.
- Conducted what-if and budget-allocation scenarios to evaluate the impact of shifting marketing spend across channels and support upcoming brand planning cycles.
- Translated model outputs into actionable promotional investment recommendations for pharma brand teams, balancing model performance, interpretability, and business constraints.

Clinical Trial Patient Enrollment Forecasting

- Developed an XGBoost regression model to forecast 6-month patient enrollment at clinical trial sites across approximately 8,000 site-trial records from 90+ historical studies, enabling data-driven site selection and recruitment planning.
- Engineered 40+ site-, trial-, therapeutic-area-, and country-level features from historical recruitment performance, trial characteristics, site experience, and operational factors; implemented time-aware validation with study-level holdouts to prevent leakage across sites sharing a protocol.
- Benchmarked against historical-average, linear, and Random Forest baselines with cross-validated hyperparameter tuning, reducing MAE from 2.8 to 2.1 patients (25% improvement, R^2 0.58) over the incumbent planning approach.

ZS Associates — *Associate Consultant*

Nov 2020 – May 2023

340B Program Data Framework and HCP Verification Automation

- Designed a 340B data framework integrating multiple retail and non-retail datasets into a Databricks and PySpark backend, providing a single source of truth for program reporting.
- Built a web-scraping pipeline gathering HCP information from hospital websites and the NPPES registry, eliminating substantial manual verification effort.
- Trained an NER model to extract HCP names, degrees, and addresses from unstructured text; applied fuzzy matching to establish HCP-HCO affiliations and link child accounts to parent clinics.
- Automated performance reporting for contracted clinics, replacing recurring manual reporting cycles.

Automated Data Pipelines for Power BI Reporting

- Designed and implemented data pipelines aligned with business requirements to enable automated refresh of multiple Power BI dashboards used by client stakeholders.

Analytics Software Implementation and Cloud Migration

- Led offshore delivery for an analytics software rollout integrating 30+ source systems, including migration of legacy workloads to the cloud.
- Coordinated 15+ stakeholders across vendors and internal teams; managed cross-workstream dependencies, RAID logs, and executive status reporting.

Divgi-TTS — *Data Analyst Intern, Automotive*

Jun 2019 – May 2020

- Performed data analytics and statistical hypothesis testing in Python to support data-driven decision-making across automotive manufacturing workflows.

EDUCATION

M.Tech, Project Management — College of Engineering Pune (COEP) | CGPA 9.3/10

2020

B.E., Mechanical Engineering — Savitribai Phule Pune University (SPPU) | 77%

2017

CERTIFICATIONS

- Databricks Accredited Generative AI Fundamentals
- Databricks: Large Language Models (LLMs) — Application through Production
- Deep Dive in Data Science with AI and ML — IVY Professional School

AWARDS AND RECOGNITION

- BRAVO Award (Q4 2023), Axtia — for impact delivery on a strategic data science engagement.
- In-Genius Award (2022), ZS Associates — for an Excel-based planning tool that streamlined a recurring business workflow.
- Project Champion and PMEC Excellence Award (2021), ZS Associates — recognising end-to-end project leadership.